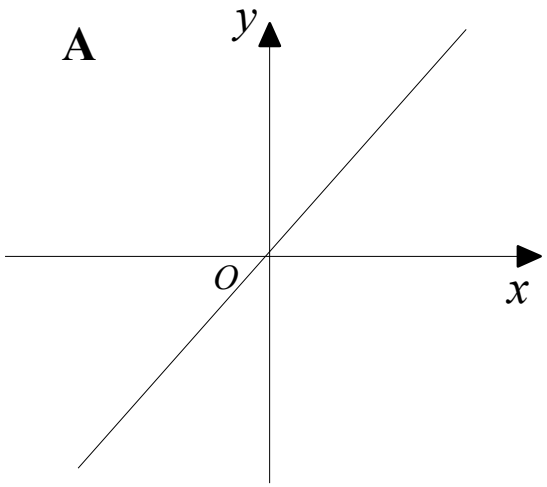
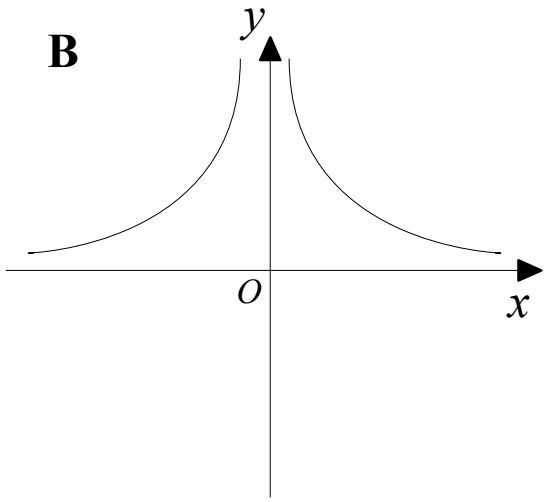
1 Here are four graphs.

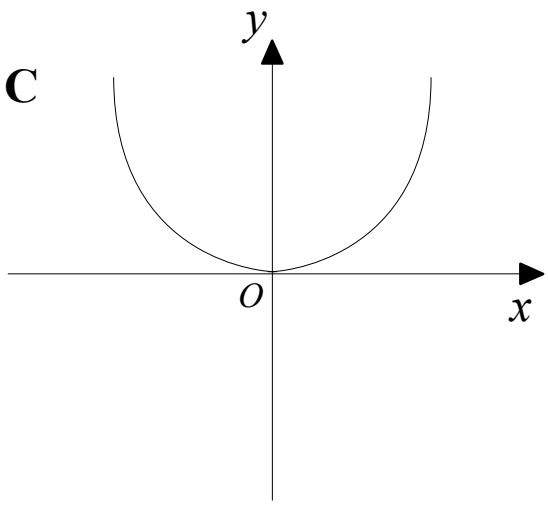
A



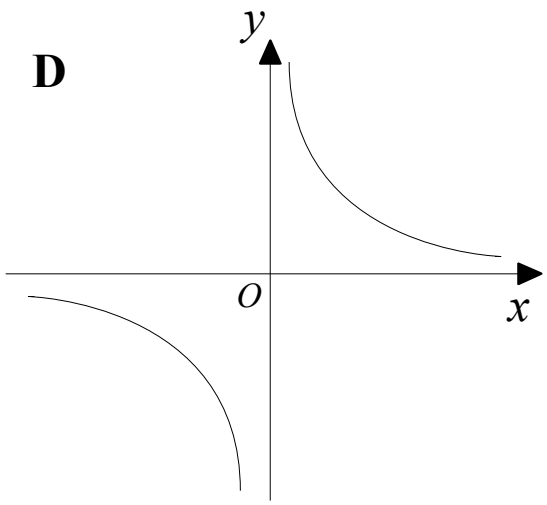
B



C



D



Match each graph with a statement in the table below.

Proportionality relationship	**Graph letter**
y is directly proportional to x	
y is inversely proportional to x	
y is directly proportional to x^2	
y is inversely proportional to x^2	

(Total for question 1 is 2 marks)

2 a is directly proportional to b

When $a = 7$, $b = 28$

Find the value of b when $a = 5$

$b =$…….....

(Total for question 2 is 3 marks)

---

3 c is inversely proportional to d

When $c = 3$, $d = 8$

Find the value of c when $d = 2$

$c =$…….....

(Total for question 3 is 3 marks)

4 e is directly proportional to f

When $e = 3$, $f = 36$

Find the value of f when $e = 4$

$f =$.....

(Total for question 4 is 3 marks)

5 g is directly proportional to the square root of h

When $g = 18$, $h = 16$

Find the possible values of h when $g = 2$

$h =$.....

(Total for question 5 is 3 marks)

6 y is inversely proportional to x

When $y = 15$, $x = 4$

Find the value of y when $x = 12$

$y =$.....

(Total for question 6 is 3 marks)

7 x is inversely proportional to the square root of y

When $x = 12$, $y = 9$

Find the value of x when $y = 81$

$x =$.....

(Total for question 7 is 3 marks)

8 y is inversely proportional to the cube of x

When $y = 250$, $x = 0.2$

Find the value of y when $x = 0.5$

$y =$.....

(Total for question 8 is 3 marks)

9 x is directly proportional to the cube of y

When $x = 32$, $y = 0.4$

Find the value of y when $x = 256$

$y =$.....

(Total for question 9 is 3 marks)

10 The table shows pairs of values for x and y

x	2	3
y	32	72

(i) Tick the correct statement below.

$y \propto x$ ….....

$y \propto x^2$ ….....

$y \propto x^3$ ….....

(ii) Write a formula for y in terms of x

….....

(Total for question 10 is 4 marks)

11 The table shows pairs of values for x and y

x	4	5
y	256	500

(i) Tick the correct statement below.

$y \propto x$ .....

$y \propto x^2$ .....

$y \propto x^3$ .....

(ii) Write a formula for y in terms of x

.....

(Total for question 11 is 4 marks)